

PPS 3.3 - Projectile Motion

1. X and Y will hit the floor at the same time (C)

2. $s_v = ut + \frac{1}{2}gt^2$

$$t = \sqrt{\frac{2 \times 80}{9.81}} = 4.04\text{s}$$

$$4.04 \times 60 = 242.4 \approx 240\text{m (C)}$$

3. $s_v = ut + \frac{1}{2}at^2$

$$t = \sqrt{\frac{2s_v}{a}} = 3.06\text{s}$$

$$s_h = ut + \frac{1}{2}at^2$$

$$= 72 \times 3.06 = 220\text{m (C)}$$

4. $30 \cos 20^\circ = 28$ (C)

5. $s = ut + \frac{1}{2}at^2$

$$t = \sqrt{\frac{2s}{a}} = 0.639\text{s}$$

$$s = ut + \frac{1}{2}at^2 = 10t = 6.4\text{m}$$

6.

a. $3 \sin 30^\circ = 1.5\text{ms}^{-1}$.

$$v_v^2 = u_v^2 + 2as$$

$$s = \frac{v_v^2 - u_v^2}{2a} = \frac{1.5^2}{2 \times 9.81} = 0.11\text{m}$$

b. $s = ut + \frac{1}{2}at^2 \Rightarrow t = \sqrt{\frac{2s}{a}}$

$$t = \sqrt{\frac{2 \times 1.5}{9.81}} = 0.553\text{s}$$

c. $v_h = 3 \cos 30^\circ = 2.60\text{ms}^{-1}$

$$s_h = ut + \frac{1}{2}at^2 = 2.60 \times 0.553 = 1.44\text{m (a = 0)}$$

7. Assuming the horizontal distance between the barrel of the gun and the launching device is 170m:

$$\text{Distance between barrel of the gun and clay pigeon} = \sqrt{170^2 + 60^2} = 180.277\text{m}$$

At 300ms^{-1} , it takes $\frac{180.277}{300} = 0.6\text{s}$ for the bullet to travel from the end of the gun's barrel to the clay pigeon.

In 0.6s the clay pigeon will have fallen $s\text{m}$, where $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 9.81 \times 0.6^2 = 1.77\text{m}$. This distance is negligible to the time taken for the bullet to hit the clay pigeon.

Therefore the time taken does not change.

8.

a. horizontal component = $63 \cos 14^\circ = 61 \text{ms}^{-1}$
vertical component = $63 \sin 14^\circ = 15 \text{ms}^{-1}$

b. $s_h = ut + \frac{1}{2}at^2$
 $= 61 \times 4.9 = 299 \text{m} \ (a = 0)$

c. $s_v = ut + \frac{1}{2}gt^2$ where $g = -9.81$
 $= 15 \times 4.9 + \frac{1}{2} \times -9.81 \times 4.9^2$
 $= -44 \text{m}$

d. $\tan \theta = \frac{44}{299}$
 $\theta = \arctan\left(\frac{44}{299}\right) = 8.4^\circ$

Extension Questions

Rope Swing

a.

We know that all of the person's vertical velocity at A becomes horizontal velocity at B .
i.e. at A , $v_h = 0$ and $v_v = v$ and at B $v_h = v$ and $v_v = 0$.

Thus the velocities can be expressed as $v_v^2 + v_h^2 = r^2$

1. Finding the time taken for the person to fall r

$$s_v = u_v t + \frac{1}{2}at^2$$

$$r = 0 + \frac{g}{2}t^2$$

$$\therefore t = \sqrt{\frac{2r}{g}}$$

2. Finding the time taken to travel from A to B .

Since energy is conserved from A to B , $\frac{1}{2}mv^2 = mgh$.

$$\therefore v = \sqrt{2gr} \ (h = r)$$

Since we know the time taken to fall r , we can use the equation

$$s = ut + \frac{1}{2}at^2$$

to determine the distance travelled.

$$\begin{aligned} s &= (\sqrt{2gr}) \left(\sqrt{\frac{2r}{g}} \right) + \frac{g}{2} \left(\sqrt{\frac{2r}{g}} \right)^2 \\ &= \sqrt{4r^2} + r \\ &= 2r + r \\ &= 3r \end{aligned}$$

Therefore, the horizontal distance d covered by the person is $3r$.

- b. If the person lets go at an angle $\theta = 45^\circ$ to the vertical then they let go when $|v_h| = |v_v|$.

We get from part a. that $v = \sqrt{2gh}$, but $h = r \cos 45^\circ$, not r .

$$\text{So } v = \sqrt{2g\left(\frac{r}{\sqrt{2}}\right)} = \sqrt{\sqrt{2}gr}.$$

We can say that $v = \sqrt{v_h^2 + v_v^2}$ where v_h and v_v are the horizontal and vertical components of v respectively. But since $\theta = 45^\circ$, $v = \sqrt{2v_v^2}$

Then

$$\begin{aligned}\sqrt{2v_v^2} &= \sqrt{\sqrt{2}gr} \\ 2v_v^2 &= \sqrt{2}gr \\ v_v &= \sqrt{\frac{gr}{\sqrt{2}}} = v_h\end{aligned}$$

Now we have (in the vertical direction)

$$\begin{aligned}s &= 2r - r \cos 45^\circ \\ u &= \sqrt{\frac{gr}{\sqrt{2}}} \\ a &= g = -9.81 \text{ms}^{-2} \\ t &= ?\end{aligned}$$

We know that $s = ut + \frac{1}{2}at^2$

So

$$\begin{aligned}2r - r \cos 45^\circ &= \left(\sqrt{\frac{gr}{\sqrt{2}}}\right)t + \frac{1}{2}gt^2 \\ \left(\frac{1}{2}g\right)t^2 + \left(\sqrt{\frac{gr}{\sqrt{2}}}\right)t - (2r - r \cos 45^\circ) &= 0\end{aligned}$$

Using the quadratic formula $t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, we get

$$\begin{aligned}t &= \frac{-\left(\sqrt{\frac{gr}{\sqrt{2}}}\right) \pm \sqrt{\left(\sqrt{\frac{gr}{\sqrt{2}}}\right)^2 - 4\left(\frac{1}{2}g\right)(2r - r \cos 45^\circ)}}{2\left(\frac{1}{2}g\right)} \\ &= \frac{-\sqrt{\frac{gr}{\sqrt{2}}} + \sqrt{\frac{3gr - 4\sqrt{2}gr}{\sqrt{2}}}}{g}\end{aligned}$$

this is definitely wrong but i literally do not know how to do it correctly. i'm giving up.